With 750 labelled tweets, does fine-tuning beat a calibrated open LLM?

RESEARCH QUESTION · UNCHANGED · FORMAL TEST: THE TWO PROPOSAL LLMS

Take **750 labelled OffensEval-TR tweets**. Fine-tune **BERTurk**, or prompt **Gemma-2-9B-it** or **Qwen3-8B** with 8 examples from those 750 and use the other 742 to set a decision threshold. Compare **macro-F1** on the frozen 3,528-tweet test set.

WE CALL IT A NO WHEN

BERTurk's mean over 10 runs is less than **0.02** above the LLM's mean over 3 runs, or a paired bootstrap says the gap could be chance.

Applies to Gemma-2-9B and Qwen3-8B only. Qwen3-14B and Gemma-4-12B were added after the first dev results and are reported as extra analysis.

0.613	0.767	0.790	60
TF-IDF-750	BERTurk-750	Gemma-4-12B	baseline errors
dev, 10 runs	dev, 10 runs	dev, extra analysis	audited

Vesile Han · Gürkan Göztepeli (CSE 654, research leads) · Mert Uğur · Utku Gökçek (CSE 484)

30,000 train / 1,756 dev / 3,528 frozen test; 19.3% offensive

FILE / SPLIT	TWEETS	NOT	OFF	OFF %
Official training file	31,756	25,625	6,131	19.3
Our train (seed 42)	30,000	24,208	5,792	19.3
Our dev	1,756	1,417	339	19.3
Official test (frozen)	3,528	2,812	716	20.3

OffensEval-TR 2020 (Çöltekin, LREC 2020). Annotations CC BY 2.0. @USER/URL already masked; no extra cleaning. 9 duplicate train texts; 1 train/test overlap.

REAL EXAMPLE · OFF · TRAIN ID 33518

“Kırmızı ağır karar Allah belanı versin”

“A red card is a harsh call, God damn you.” A curse aimed at a person (“you”, likely the referee): OFF.

CONTRAST · NOT

“...annem alacağı arabadan kaç kere vazgeçti ya kahretsin zalım trafik”

“How many times my mum gave up on buying a car, damn this cruel traffic.” Cursing at traffic is not offensive; target matters.

WHY THESE COUNTS MATTER

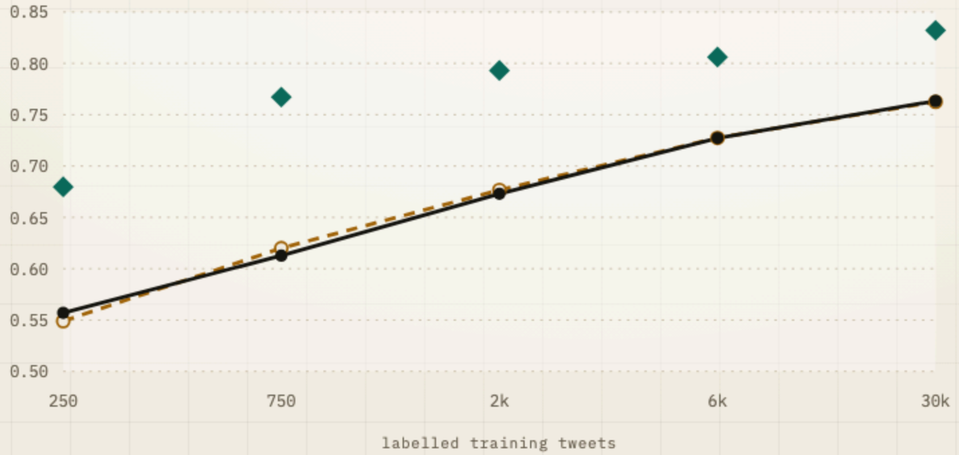
Only 19.3% of training tweets are offensive. A model that only answers “not offensive” looks right at first glance: 80.7% accuracy. Its macro-F1 is 0.447, though, so accuracy alone hides the failure. The test set stays frozen; dev is only for code checks and mid-project numbers, and every system uses the saved subsets (seed 42 + k).

TF-IDF + LR reaches 0.763 on test; 0.620 with 750 labels

SYSTEM	TEST MACRO-F1	OFF P	OFF R	SPREAD
Always NOT	0.444	n/a	0.000	deterministic
TF-IDF + LR, 750	0.620 ± 0.012	n/a	n/a	sd over 10 samples
TF-IDF + LR, 30,000	0.763	0.589	0.673	95% CI [0.745, 0.779]

The frozen test set was scored once for this deterministic baseline. All method results stay on dev until the final freeze.

0.620 test at 750 labels
0.613 on dev



Solid: TF-IDF dev. Dashed: TF-IDF test. Diamonds: fine-tuned BERTurk dev.

WHERE THE FLOOR SITS

With 750 labels, the simple baseline scores 0.620 on test and 0.613 on dev. That is the minimum BERTurk and the LLMs must beat. At 30,000 labels, TF-IDF climbs to 0.763; BERTurk scores 0.767 with only 750 labels and 0.832 with 30,000 on dev.

275 dev errors; a 60-error audit shows where a word model fails

FALSE ALARMS (166): CATEGORY	N
No insult; spurious word/char cue	9
Political or religious criticism	10
Sports criticism or banter	5
Personal criticism not judged offensive	3
Sarcasm or irony	1
Label noise or genuinely ambiguous	2

MISSED OFFENSIVE (109): CATEGORY	N
Implicit insult, no profanity	8
Masked or abbreviated profanity	5
Veiled threat or violence	3
Political or religious insult	4
Institutional or sports criticism	3
Label noise or not offensive	7

Counts are in the 60-error audit (30 per direction, seed 42); each table sums to 30.

FALSE ALARM · F3 · 13105

"Aziz Yıldırım Fenerbahçe'yi padişahlık zannediyor herhalde..."

Club criticism without a slur.

MISS · M1 · 43325

"...bundan daha parazit canlı türü yok"

Implicit insult with no swear word.

MISS · M2 · 15069

"...bakanlık seçimi kaldı amkk"

Masked profanity, unseen by the word/char model.

WHAT THE AUDIT SUGGESTS

False alarms often involve conflict and club words and character combinations. Implicit insults and masked profanity are easily missed. About one in seven audited errors is label noise, so the ceiling is below 1.0.

Both arms run end-to-end; first dev numbers

CONTROLLED CHANGE

The baseline trains its weights on the 750 labels. The LLM uses those same 750 labels to pick 8 examples and a threshold. Nothing else differs; only how the labels are used.

LLM ARM

750 labels → 8 prompt examples (4 OFF / 4 NOT) → 742 labels set the threshold → score dev as log P(A) − log P(B).

FINE-TUNING ARM

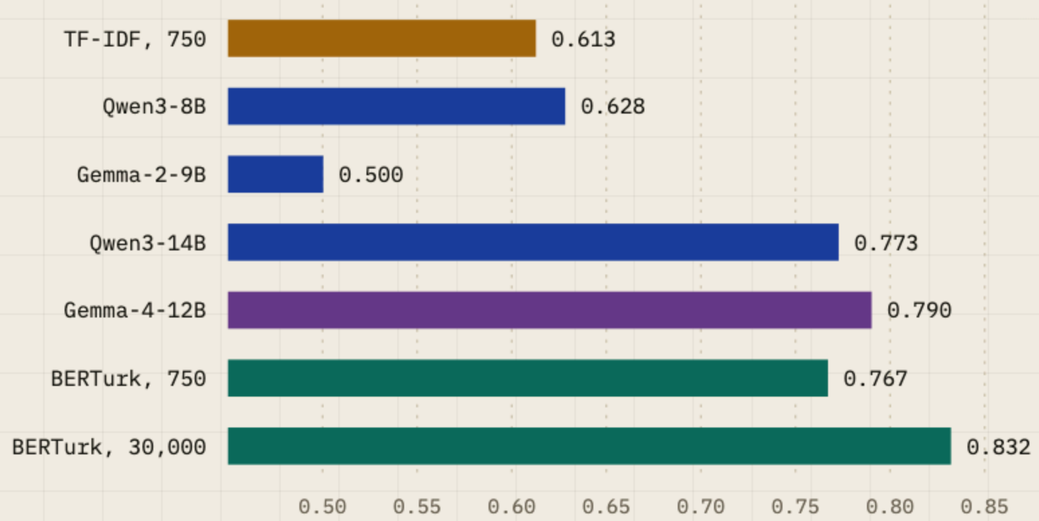
BERTurk and XLM-R-base, same subsets: 128 tokens, batch 16, lr 2e-5, 4 epochs or 400 updates, no early stopping. 46 runs.

SYSTEM (DEV)	MACRO-F1	RUNS	THRESHOLD
Qwen3-8B, zero-shot	0.162	1	0
Qwen3-8B, 8 + 742	0.628 ± 0.034	3	3.13 / 2.75 / 2.50
Gemma-2-9B, zero-shot	0.412	1	0
Gemma-2-9B, 8 + 742	0.500 ± 0.048	3	0.63 / 0.25 / 0.88
BERTurk, 750	0.767 ± 0.014	10	n/a
BERTurk, 30,000	0.832 ± 0.003	3	n/a
XLM-R-base, 750 (Ablation A)	0.670 ± 0.025	10	n/a

READING THE DEV TABLE

Zero-shot to calibrated is +0.466 macro-F1 for Qwen3-8B and +0.088 for Gemma-2-9B. BERTurk-750 sits at 0.767, +0.097 above XLM-R-base on the same samples (Ablation A, 95% CI 0.080 to 0.116). These are dev numbers; the test comparison is still pending.

BERTurk beats both proposal LLMs; Gemma-4-12B leads on dev



SYSTEM	DEV MACRO-F1	RUNS
TF-IDF + LR, 750	0.613 ± 0.015	10
Qwen3-8B	0.628 ± 0.034	3
Gemma-2-9B	0.500 ± 0.048	3
Qwen3-14B	0.773 ± 0.009	3
Gemma-4-12B	0.790 ± 0.007	3
BERTurk, 750	0.767 ± 0.014	10
BERTurk, 30,000	0.832 ± 0.003	3

+0.154
BERTurk-750 vs TF-IDF

+0.139
vs Qwen3-8B

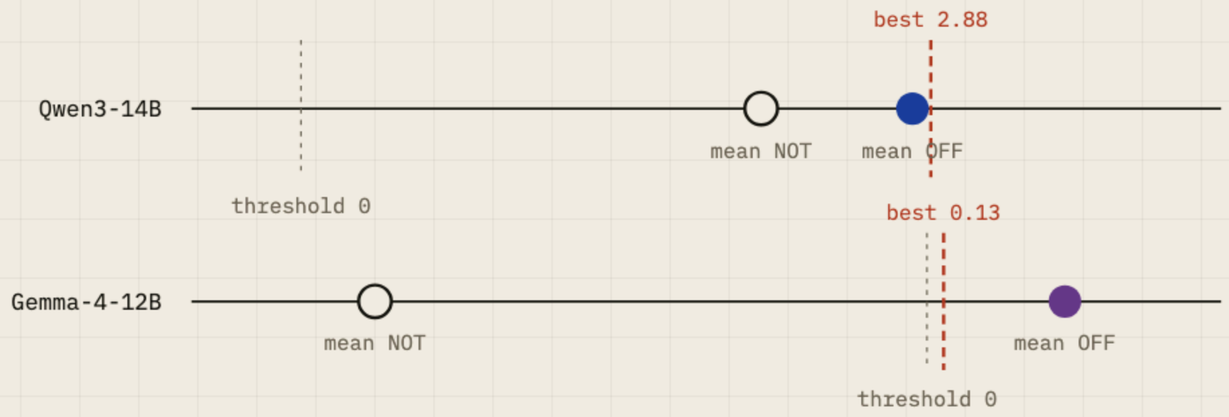
+0.267
vs Gemma-2-9B

−0.023
Gemma-4-12B ahead of BERTurk

Dev only. The frozen-test comparison is still pending.

Qwen is shifted; Gemma's threshold is already near optimal

ZERO-SHOT DIAGNOSTIC	F1 @ 0	F1 @ BEST (ORACLE)	AUC	MEAN OFF	MEAN NOT
Qwen3-14B	0.171	0.677	0.756	2.79	2.10
Gemma-4-12B	0.763	0.765	0.864	1.03	-4.13



Mean log P(A) − log P(B); zero and best thresholds marked. F1 @ 0 uses threshold 0.

WHAT EXPLAINS THE GAP

- **It is not the tokenizer.** Gemma packs these tweets more tightly: 32.3 vs 37.9 tokens/tweet.
- **Qwen is shifted, not blind.** At threshold 0 it calls 99% of tweets offensive, but the ranking still works: best-threshold F1 is 0.677.
- **Gemma is already centred.** F1 at 0 is 0.763 and the best threshold reaches 0.765, so threshold tuning has almost nothing to gain here.
- **Oracle, diagnostic only.** F1 @ best picks the threshold on dev itself, so it is an upper bound, not a result; no setting is chosen from it.

This is threshold geometry, not proof of contamination. It does not separate pretraining language from alignment; the exact-completion check is still planned.

One dev pass on an L4: 3 to 5 tweets/s with examples, 10 to 18 without

SYSTEM	PARAMS	DEV SCORING	THROUGHPUT	PEAK VRAM	NOTES
TF-IDF, 750	n/a	0.6 s	~3,100/s	CPU	fit 1.0 s
TF-IDF, 30,000	n/a	0.6 s	~3,100/s	CPU	fit 14.6 s
Qwen3-8B, few-shot	8.2B	355 s	5.0/s	n/a	3 runs
Gemma-2-9B, few-shot	9.2B	430 s	4.1/s	n/a	3 runs
Qwen3-14B, few-shot	14.8B	644 s	2.7/s	n/a	3 runs
Gemma-4-12B, few-shot	12.0B	579 s	3.0/s	n/a	3 runs
Qwen3-14B, zero-shot	14.8B	180 s	9.8/s	10.7 GB	load 157 s · 1,290 tok/s
BERTurk, 750 (fine-tuned)	110.6M	1.1 s	~1,550/s	2.6 GB	train 36 s
XLM-R-base, 750	278.0M	1.2 s	~1,440/s	5.2 GB	train 64 s

Scoring the 1,756 dev tweets; model download and load are excluded unless stated. Few-shot is 3 to 4× slower because each prompt carries 8 examples (~600 tokens). BERTurk scores dev 300 to 500× faster than few-shot LLMs; all 46 fine-tuning runs cost about 3 Colab units.

BERTurk/XLM-R: run 1 at 750 re-measured; VRAM is the training peak; 30,000 labels ~11 min alone. Gemma-4-26B-A4B did not fit (22 GB). 9.4 Colab units. Proposal said free T4; we used paid Colab Pro L4, for speed.

Done: BERTurk and Ablation A. Every remaining step has a date and an owner

DATE	EXPERIMENT	OWNER	STATUS
11 Nov	BERTurk 750 (10) + 30,000 (3) on dev	Vesile · Mert	done
11 Nov	Ablation A: BERTurk vs XLM-R, 750	Vesile	done
12 Nov	Mid-project talk; freeze this deck	All	talk
21 to 24 Nov	Ablation B: Gemma with English instructions	Gürkan	planned
25 to 27 Nov	Test-contamination check, first 200 test tweets	Gürkan · Utku	planned
28 to 30 Nov	Freeze methods; open test; final runs	All	planned
1 to 5 Dec	Paired bootstrap and error analysis	Mert · Utku	planned
1 to 5 Dec	Paper draft	All	planned
8 Dec	Final report	All	planned

BIGGEST RISK

A 26B+ model may not fit the L4: Gemma-4-26B-A4B OOM'd at 22 GB, so we used the 12B dense model. **Fallback:** Qwen3-14B alone.

RESULT RISK

The lines may never cross. The 750-label question still has a clear answer; the learning curve is descriptive.

On proposal feedback: the rubric asks for test-set baseline numbers, so we scored the deterministic TF-IDF baseline once on test and kept methods on dev.

RULE

Methods freeze before the test set opens. The final answer uses the paired bootstrap; the learning curve stays descriptive.

What we know, what we do not, and what comes next

BASELINE · 750 LABELS

0.613

TF-IDF dev macro-F1 with the same 750 labels (10 runs); 0.620 ± 0.012 on the frozen test set. With 30,000 labels it reaches 0.763 on test (95% CI [0.745, 0.779]).

BEST LLM · DEV · EXTRA ANALYSIS

0.790

Gemma-4-12B dev macro-F1 after calibration; zero-shot 0.763. Qwen3-14B reaches 0.773. Threshold from 742 labels; 3 runs.

FINE-TUNE · DEV

0.767

BERTurk-750 dev (10 runs); 30,000 labels reaches 0.832. It beats Qwen3-8B and Gemma-2-9B, the two LLMs in the formal test. Gemma-4-12B (extra analysis) leads on dev by 0.023.

Still missing: test-set scores for both arms. Everything above is dev; the final verdict uses the frozen 3,528-tweet test set. Four decisions were fixed before any method ran: the test split, macro-F1, the 0.02 minimum gap, and a paired bootstrap over 10,000 resamples to check that the gap is not chance.

Questions welcome; any member can answer any part.

Title, abstract, research question, contributions

1 · TITLE AND ABSTRACT

Fine-tuning BERTurk vs threshold-calibrated open LLMs for Turkish offensive language with 750 labels.

Small Turkish platforms can label only a few hundred tweets. We ask a narrow question: with 750 labelled OffensEval-TR tweets, does a fine-tuned BERTurk beat an open LLM that gets the same 750 labels as 8 prompt examples and 742 calibration tweets for its decision threshold? The TF-IDF + logistic-regression baseline scores 0.763 macro-F1 on the frozen 3,528-tweet test set, but only 0.620 with 750 labels. The LLM arm scores each tweet as log P(A) − log P(B), and the 742 labels set its threshold. On dev, calibration moves Qwen3-8B from 0.162 to 0.628 and Gemma-2-9B from 0.412 to 0.500; Qwen3-14B reaches 0.773 (zero-shot 0.171) and Gemma-4-12B 0.790 ± 0.007 (zero-shot 0.763). Fine-tuned BERTurk reaches 0.767 ± 0.014 on dev with 750 labels. **Gap:** the frozen-test comparison is pending; the final result uses the paired bootstrap.

2 · RESEARCH QUESTION

Unchanged. With 750 labels, does fine-tuning BERTurk beat prompting Gemma-2-9B-it or Qwen3-8B (same 750 tweets, 8 examples plus 742 threshold labels) on macro-F1 over the official 3,528-tweet test set?

"No" result: BERTurk's mean over 10 runs is not ≥0.02 above the LLM's mean over 3 runs, or the paired bootstrap says chance ($p \geq 0.05$). **Change:** Qwen3-14B and Gemma-4-12B were added after the first dev results; the formal test (0.02 gap, paired bootstrap) uses only Gemma-2-9B and Qwen3-8B, and the new models are extra analysis. Compute: paid Colab Pro L4 instead of free T4.

3 · CONTRIBUTIONS

C1. At 750 labels, we test whether BERTurk's mean test macro-F1 is ≥0.02 above each LLM's, and whether a paired bootstrap rejects chance. Test pending; on dev BERTurk 0.767 vs 0.628 / 0.500.

C2. The 742-label threshold improves each LLM's dev macro-F1 over zero-shot; we report per-model delta and spread.

C3. The learning curve tests how many labels fine-tuning needs to overtake the best calibrated LLM; on dev BERTurk passes Gemma-4-12B (0.790) between 750 and 2,000 labels (0.767 to 0.793). If the lines never cross on test, the 750-label answer stands.

Related work and data

4 · RELATED WORK

1. **Zampieri et al. 2020.** "SemEval-2020 Task 12: Multilingual Offensive Language Identification in Social Media (OffensEval 2020)." SemEval. Top Turkish systems fine-tune on full data; no label-budget comparison.
2. **Çöltekin 2020.** "A Corpus of Turkish Offensive Language." LREC. The dataset; no fine-tuning-vs-prompting comparison.
3. **Zampieri et al. 2023.** "OffensEval 2023: Offensive Language Identification in the Age of Large Language Models." NLE. Zero-shot LLMs; Turkish only Flan-T5-large 0.451; no examples.
4. **Dehghan & Yanıkoğlu 2024.** "Evaluating ChatGPT's Ability to Detect Hate Speech in Turkish Tweets." CASE, pp. 54-59. ChatGPT with 7 examples vs BERTurk, one dataset and one size.
5. **Pecher, Srba & Bielikova 2025.** "Comparing Specialised Small and General Large Language Models on Text Classification." EMNLP, pp. 165-184. English break-even near 100 labels; we test noisy Turkish two-class data.
6. **Schmitt & Schweter 2025.** "SindBERT." arXiv:2510.21364. Turkish encoders, full data only; no LLMs or small budgets.
7. **Şeker, Uludoğan, Önal & Özgür 2025.** "HATECAT-TR." Findings of EMNLP. Turkish span hate; BERTurk 0.59 vs LLaMA-3.3-70B 0.53; we classify whole tweets.
8. **Karakaş & Şimşek 2026.** arXiv:2606.29614. Zero-shot LLMs vs BERTurk at one size; we add examples, calibration and a curve.

5 · DATA

OffensEval-TR 2020, Çöltekin, LREC 2020; annotations CC BY 2.0; Turkish; @USER/URL masked. No extra cleaning.

SPLIT	TWEETS	OFF	OFF %
Official train	31,756	6,131	19.3
Our train	30,000	5,792	19.3
Dev	1,756	339	19.3
Test (frozen)	3,528	716	20.3

9 duplicate train texts; 1 test/train overlap. Example (OFF, id 33518): "Kırmızı ağır karar Allah belanı versin".

Method, protocol, results so far, error analysis

6 · METHOD AND BASELINE

TF-IDF word 1-2 + char_wb 2-5, min_df=2, sublinear, Turkish lowercasing; LR C=1.0, balanced. Test 0.763 [0.745, 0.779].

Fine-tuning: BERTurk/XLM-R, 128 tokens, batch 16, 2e-5, 4 epochs or 400 updates, class-weighted, no early stopping. LLM: Qwen/Gemma 4-bit, 8 examples, 742-label threshold. One change: labels train weights vs choose examples and threshold.

Compute used: about 3 Colab units for the 46 fine-tuning runs + 2.5 L4-h for 16 LLM passes; about 18 GPU-h needed for the final. Change: paid Colab Pro L4 instead of the proposal's free T4, for speed.

7 · EVALUATION PROTOCOL

Macro-F1; frozen 3,528-tweet test; 10 BERTurk runs at 250/750/2,000 and 3 at 6,000/30,000; 3 runs per LLM at 750. Paired bootstrap, 10,000 resamples; real gap ≥0.02 and wins on ≥95%. Fixed before method results; dev for code checks only.

9 · ERROR ANALYSIS: 60 OF 275 DEV ERRORS (30 PER DIRECTION, SEED 42)

FALSE ALARMS (166)	N	EXAMPLE (DEV ID)
No insult, spurious cue	9	12420 fan poll
Political/religious	10	35124 political accusation
Sports criticism	5	13105 "...padişahlık zannediyor"
Personal criticism	3	22409 "...fake hesapları sevmiyorum"
Sarcasm	1	28737 "Ha bi sen eksiktin"
Label noise	2	33218 "Seni cehennem paklar"

MISSED OFFENSIVE (109)	N	EXAMPLE (DEV ID)
Implicit insult	8	43325 "...parazit canlı türü yok"
Masked profanity	5	15069 "...kaldı amkk"
Veiled threat	3	35789 "...idam edicem"
Political/religious	4	25929 "Gerici yobazlar..."
Institutional/sports	3	20333 "Ösym eğer kontrol ediyorsan..."
Label noise	7	10001 "...veteriner okumaya..."

8 · RESULTS SO FAR (MEAN ± SD)

SYSTEM	DEV	TEST
Always NOT	0.447	0.444
TF-IDF, 750	0.613 ± 0.015	0.620 ± 0.012
TF-IDF, 30,000	0.763	0.763
Qwen3-8B zero	0.162	n/a
Qwen3-8B cal	0.628 ± 0.034	n/a
Qwen3-14B zero	0.171	n/a
Qwen3-14B cal	0.773 ± 0.009	n/a

SYSTEM	DEV	TEST
Gemma-2-9B zero	0.412	n/a
Gemma-2-9B cal	0.500 ± 0.048	n/a
Gemma-4-12B zero	0.763	n/a
Gemma-4-12B cal	0.790 ± 0.007	n/a
BERTurk, 250	0.680 ± 0.052	n/a
BERTurk, 750	0.767 ± 0.014	n/a
BERTurk, 2,000	0.793 ± 0.006	n/a
BERTurk, 6,000	0.806 ± 0.004	n/a
BERTurk, 30,000	0.832 ± 0.003	n/a
XLM-R, 750	0.670 ± 0.025	n/a

TF-IDF at 30,000 labels is deterministic; its spread is the 95% bootstrap CI: dev [0.739, 0.787], test [0.745, 0.779]. Method test scores wait for the freeze.

Research lead, plan and risk, reproducibility

10 · RESEARCH LEAD SECTION

Vesile Han (CSE 654), fine-tuning. Pecher et al., EMNLP 2025: English break-even near 100 labels, more for two-class tasks and variance; we test Turkish noisy two-class. Schmitt & Schweter, SindBERT 2025: Turkish encoders, full data only; no LLMs. Ablation A (done): BERTurk 0.767 vs XLM-R 0.670, +0.097 [0.080, 0.116], 100% wins. KUISAIL published BERTurk 0.816 on test; our 30k runs score 0.832 ± 0.003 on dev, test after the freeze.

Gürkan Göztepeli (CSE 654), prompting. Şeker et al., HATECAT-TR 2025: BERTurk 0.59 vs LLaMA-3.3-70B 0.53 on Turkish hate spans; we classify whole tweets and score probabilities. Karakaş & Şimşek 2026: zero-shot LLMs vs BERTurk at one size; we add examples, calibration, curve. Ablation B: Gemma English instructions, 3 runs.

11 · PLAN AND RISK

Done: BERTurk curve (36 runs), Ablation A (10). 21 to 24 Nov: Ablation B (Gürkan). 25 to 27 Nov: contamination check (Gürkan). 28 to 30 Nov: freeze, open test, final runs (all). 1 to 5 Dec: bootstrap and error analysis (Mert · Utku), paper draft (all). 8 Dec: report (all).

Biggest risk: 26B+ may not fit L4; Gemma-4-26B OOM'd at 22 GB, so 12B used. Fallback: Qwen3-14B alone. **Result risk:** lines may never cross; 750-label answer stands. **Proposal feedback:** add a test-set baseline; it is reported once, while methods stay on dev.

12 · REPRODUCIBILITY

Repository: github.com/paysis/offenseval-tr-midproject. One command: `bash scripts/reproduce.sh gpu`. Seeds: split 42, subsets 42+k, bootstrap 42. Versions: Python 3.13.16, scikit-learn 1.9.1, transformers 5.22.1, torch 2.14.1, bitsandbytes 4-bit NF4.

Roles: fine-tuning side, Vesile Han and Mert Uğur; prompting side, Gürkan Göztepeli and Utku Gökçek. Ablation A belongs to Vesile, Ablation B to Gürkan. **AI tools:** a coding assistant was used for code, analysis and drafting; every number is read from saved result files and checked by the team.